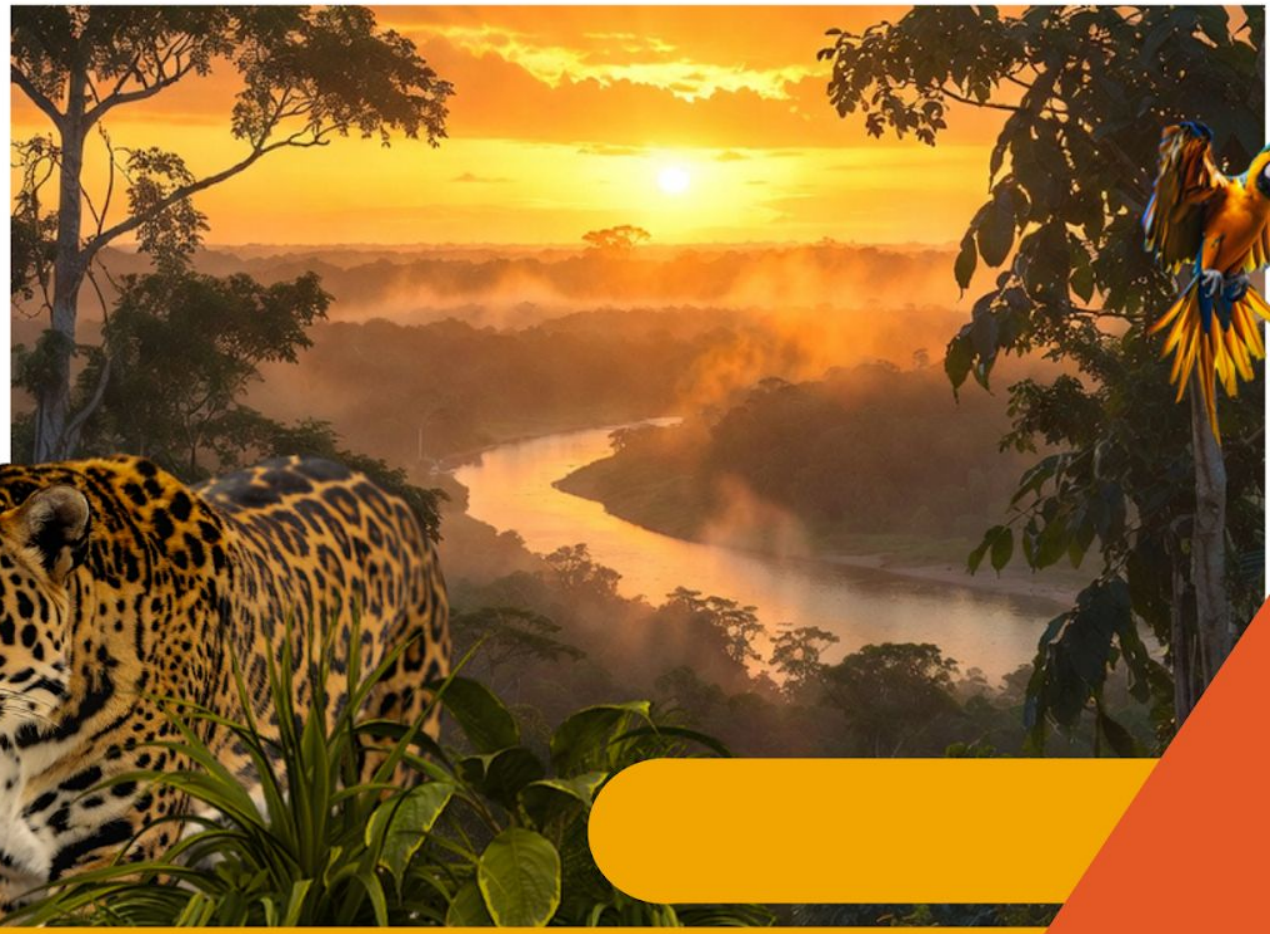
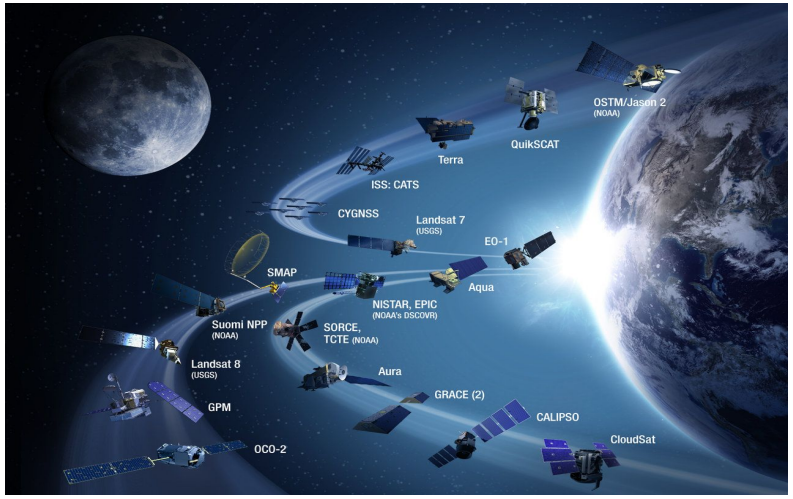
WLTS - Web Land Trajectory Service

Gabriel Sansigolo

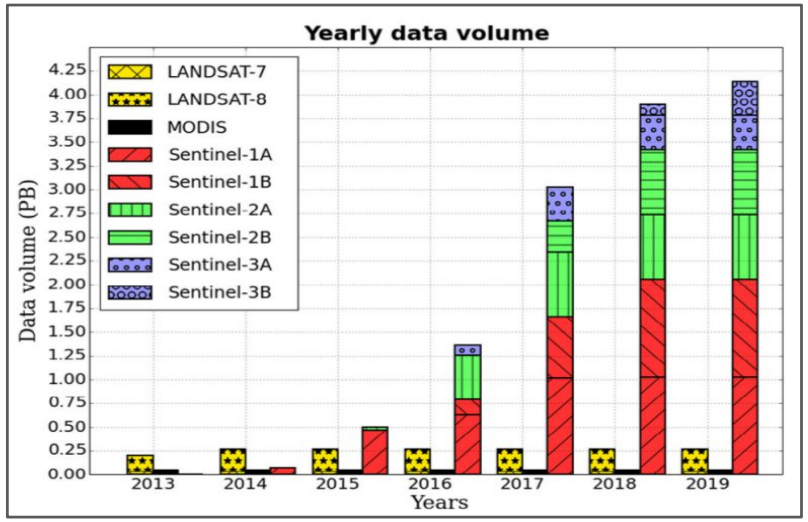
Brazilian National Institute for Space Research (INPE)



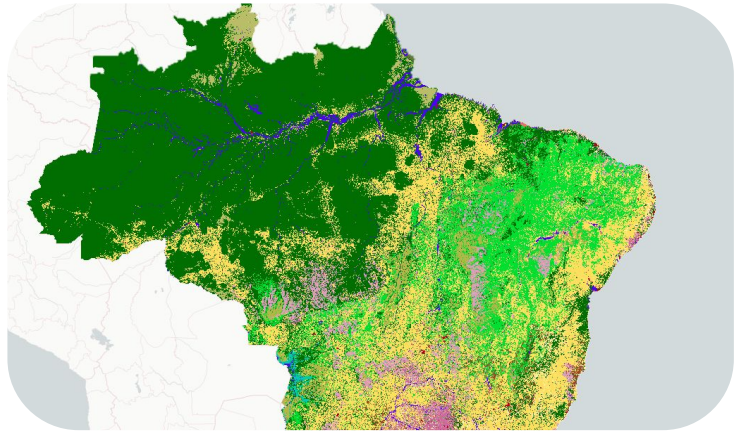
Big EO Data



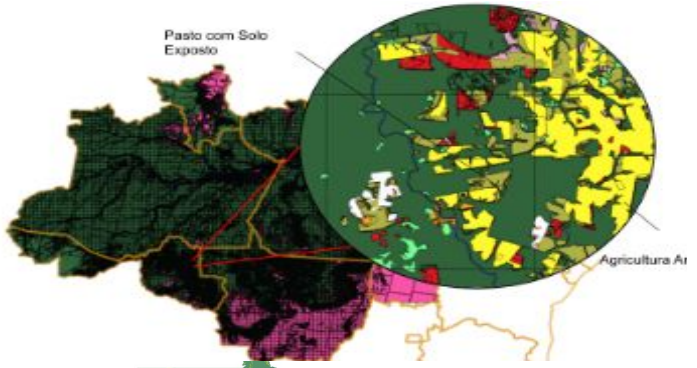
Fonte: NASA (2019)



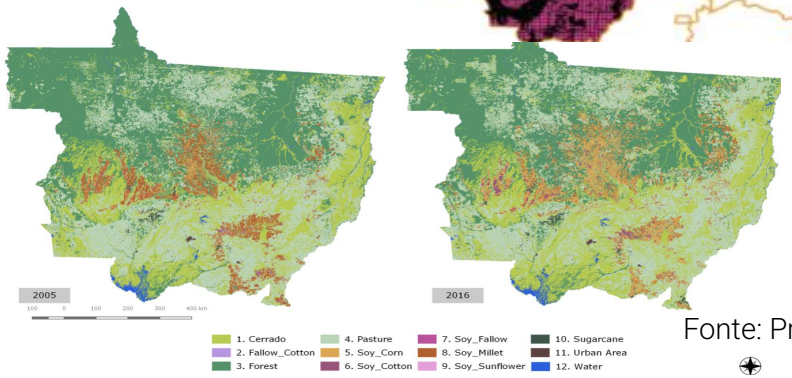
Fonte: Soille et al. (2018)



Fonte: MapBiomas



Fonte: TerraClass



Fonte: Projeto e-sensing

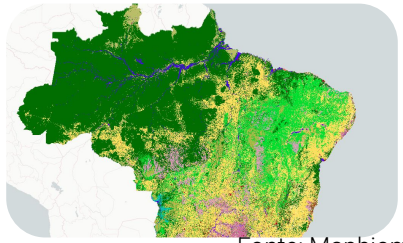
Although these projects make a large collection of data openly available, the integrated use of these collections requires a lot of effort in data collection and integration before they can be used by scientists, students, and public administrators.

PRODES

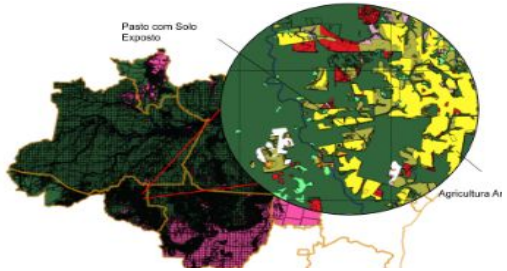
DETER

TerraClass

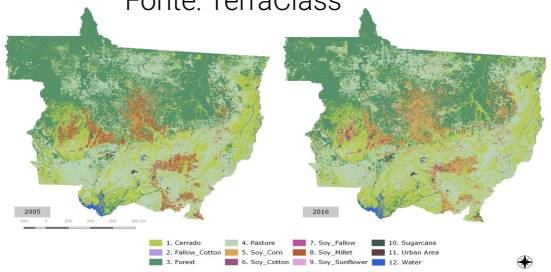
Different storage structures
(array or vector)
+
Different classification systems
+
Different temporal and spatial resolutions



Fonte: Mapbiomas



Fonte: TerraClass

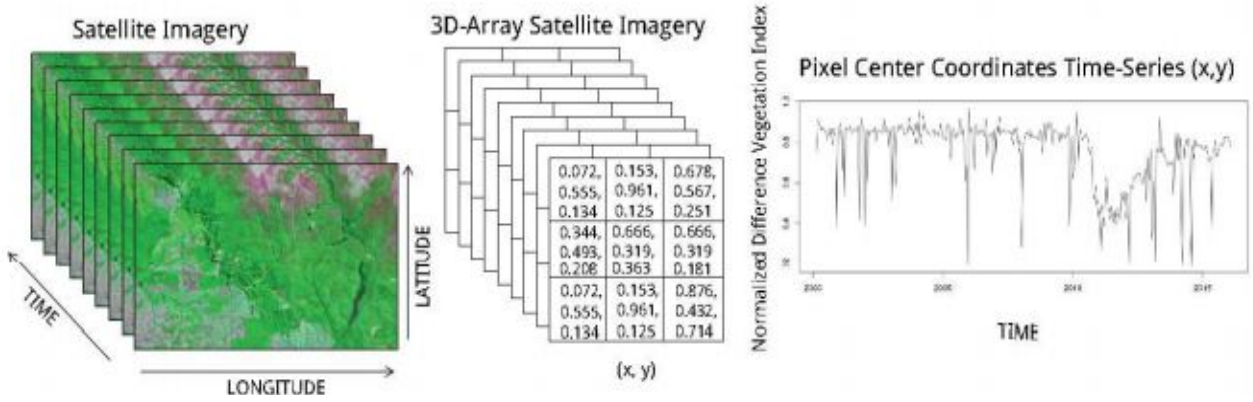


Fonte: Projeto e-sensing

Time Series

The use of **time series** derived from Earth observation satellite image **is one one of the main** research trends in the generation of classified land use and land cover maps.

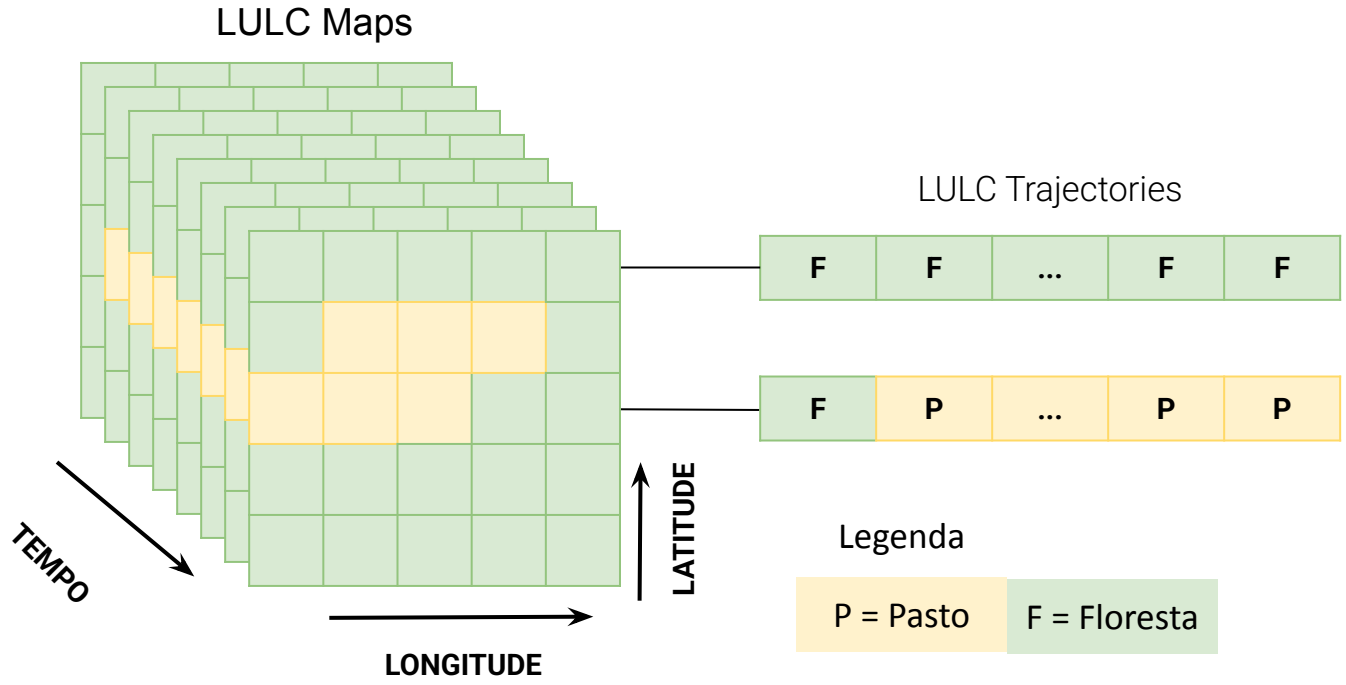
(PICOLI et al., 2018).



Fonte: Vinhas et al. (2017)

LULC Trajectories

A temporal sequence of land use or land cover classes of classified images assembled into a time series. (MENA, 2008).

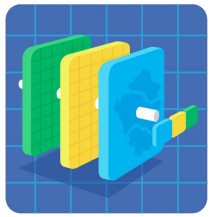


To minimize these efforts, we propose a platform that allows users to access Land Use and Land Cover (LULC) data from different sources associated with spatial locations, integrating these collections.

Specification



Services



WLTS



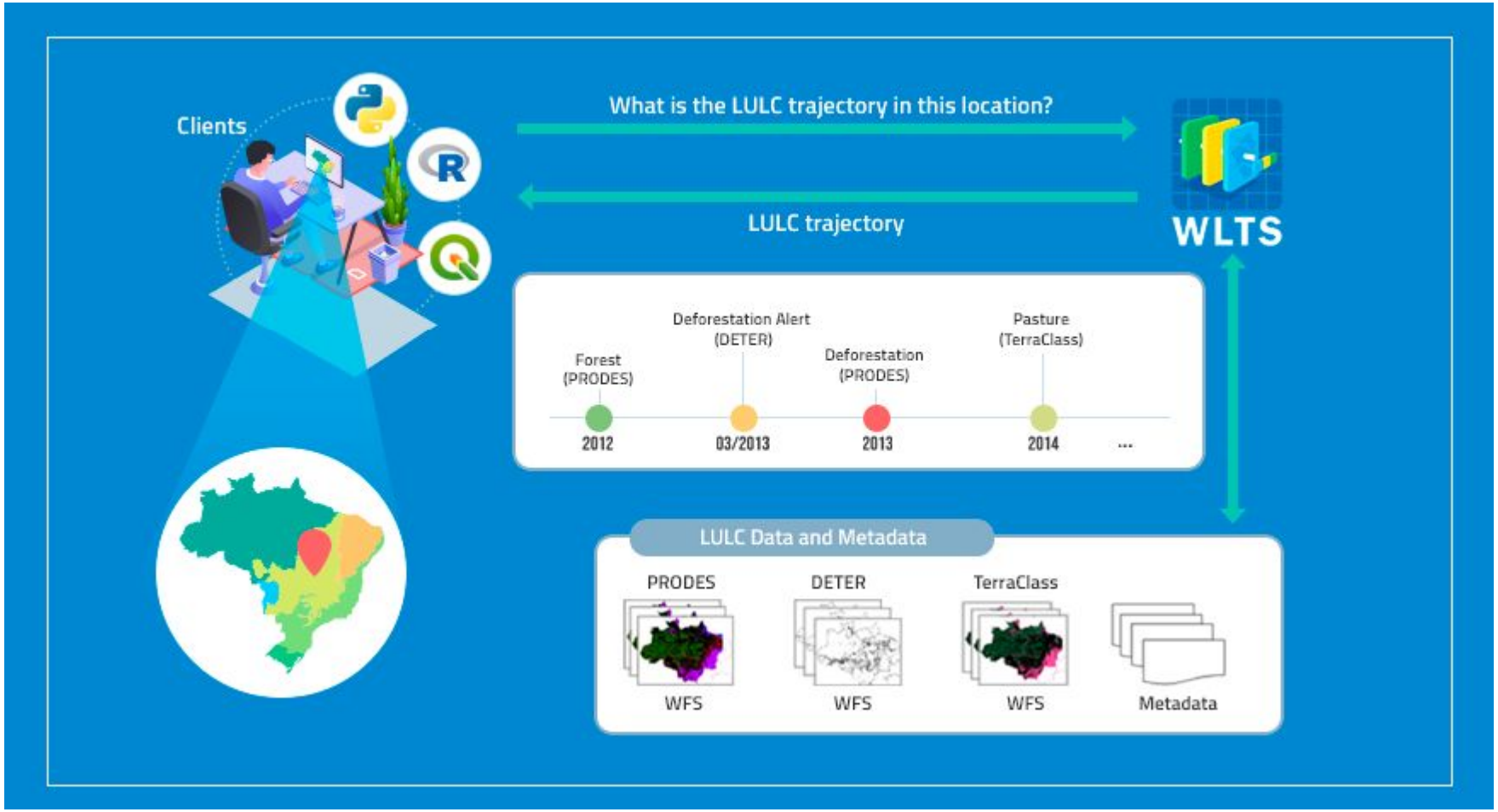
LCCS-WS

Clients

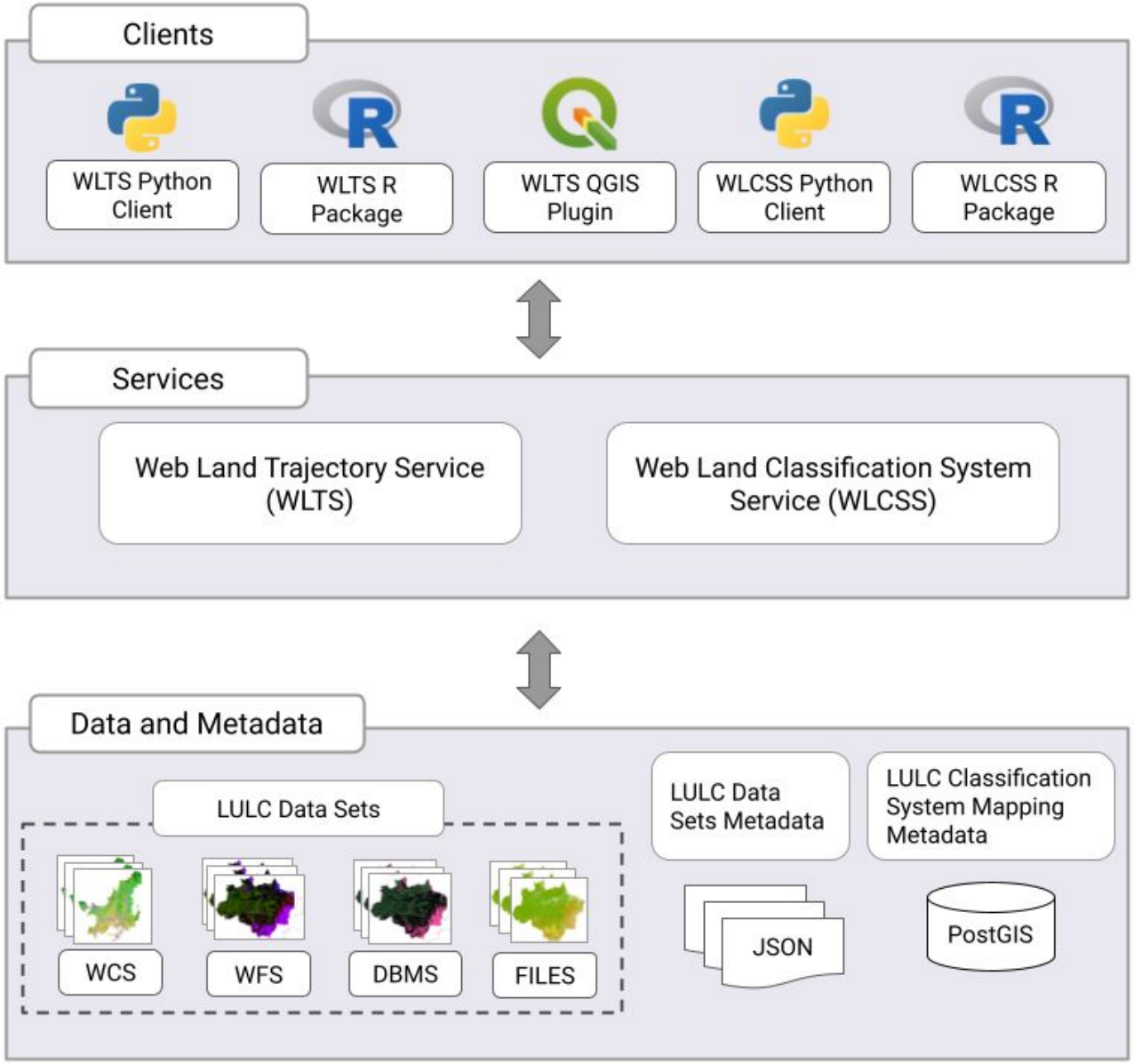




Web Land Trajectory Service (WLTS)

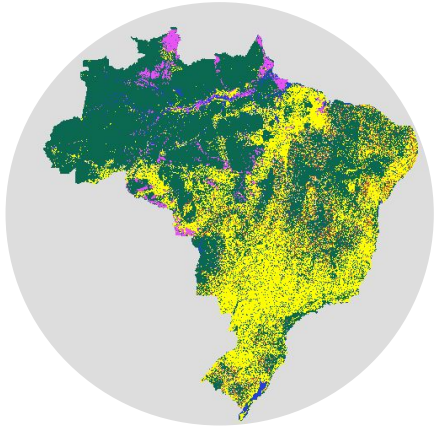


The **platform** architecture for **LULC** data integration and **trajectory** analysis.

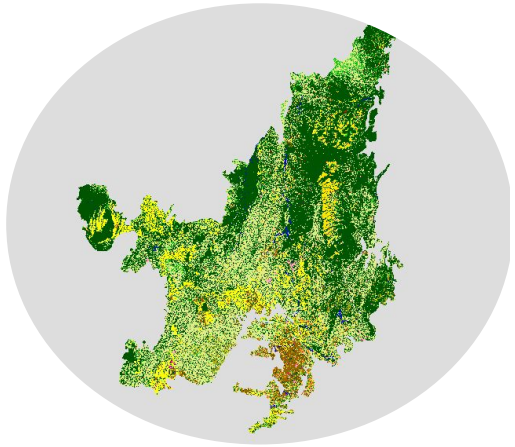


- System registered with the National Institute of Industrial Property (INPI), Process No.: BR512022003578-2

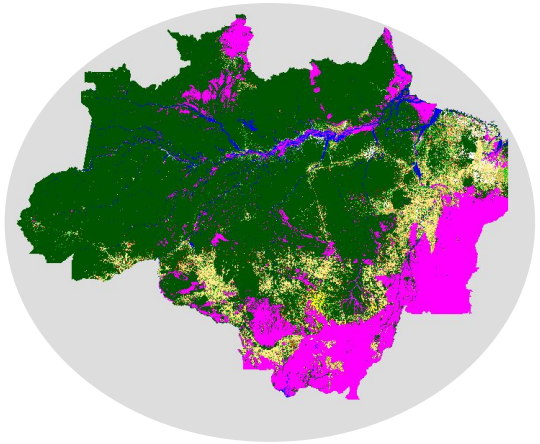
Data Collections



PRODES - Brasil
2007 - 2025* Referente a cada bioma
http://terrabrasilis.dpi.inpe.br/



TerraClass - Cerrado
2018, 2020, 2022
https://www.terraclass.gov.br/



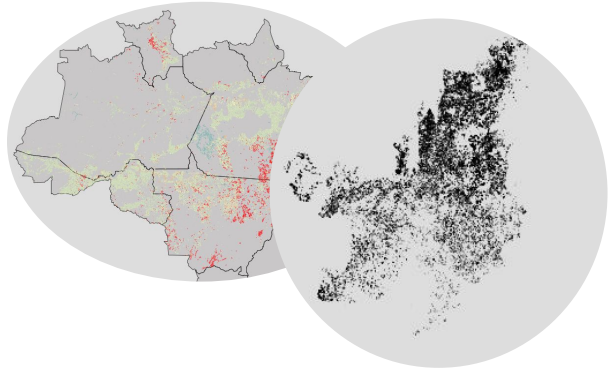
TerraClass - Amazônia
2008, 2010, 2012, 2014, 2018, 2020,2022
https://www.terraclass.gov.br/geoportal-aml/
http://terrabrasilis.dpi.inpe.br/



MapBiomas 10.0 - Brasil
1985 - 2024
https://mapbiomas.org/



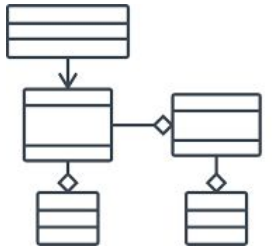
IBGE LULC - Brasil
2000, 2010, 2012, 2014, 2016, 2018, 2020



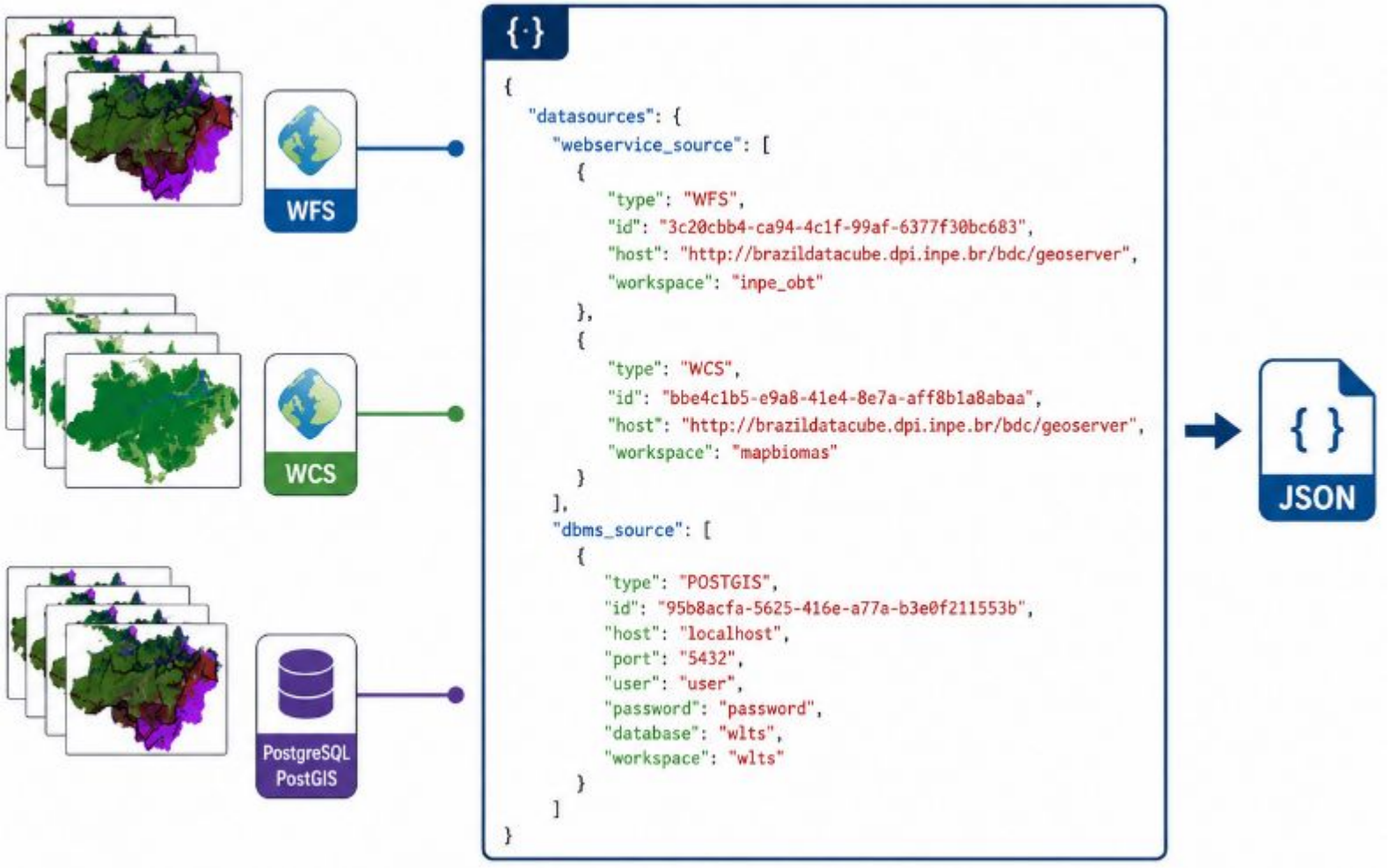
DETER - Amazônia e Cerrado
2018 - 2025
http://terrabrasilis.dpi.inpe.br/

Modelo de Integração de Dados

All metadata regarding how LULC products are stored or provided and their properties are described as JSON files.



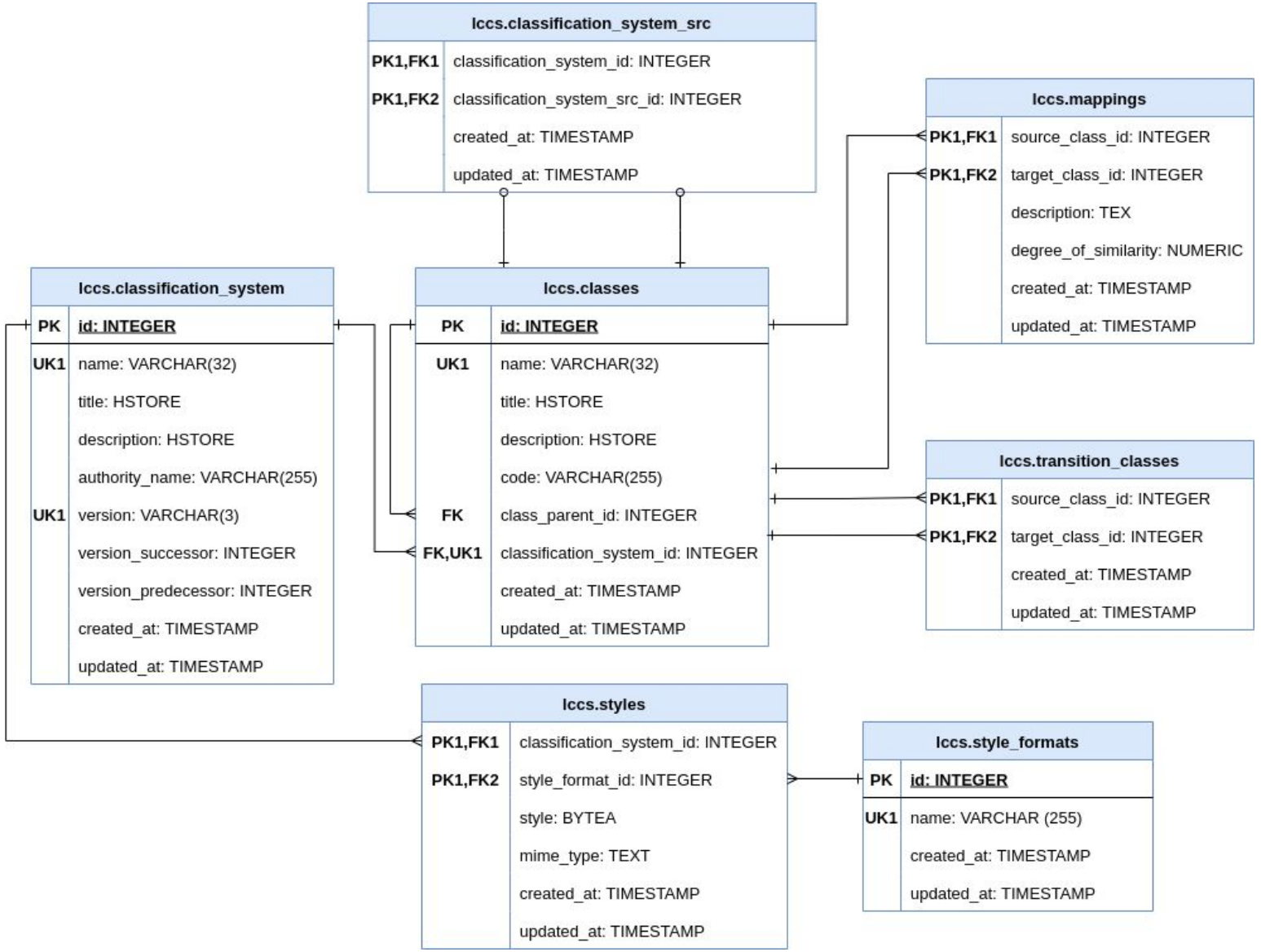
The data source types are webservice source, file sources, and dbms_sources. Each type is described as a specific set of attributes required to access LULC datasets.




```
{
  "datasources": {
    "webservice_source": [
      {
        "type": "WFS",
        "id": "3c20cbb4-ca94-4c1f-99af-6377f30bc683",
        "host": "http://brazildatacube.dpi.inpe.br/bdc/geoserver",
        "workspace": "inpe_obt"
      },
      {
        "type": "WCS",
        "id": "bbe4c1b5-e9a8-41e4-8e7a-aff8b1a8abaa",
        "host": "http://brazildatacube.dpi.inpe.br/bdc/geoserver",
        "workspace": "mapbiomas"
      }
    ],
    "dbms_source": [
      {
        "type": "POSTGIS",
        "id": "95b8acfa-5625-416e-a77a-b3e0f211553b",
        "host": "localhost",
        "port": "5432",
        "user": "user",
        "password": "password",
        "database": "wlts",
        "workspace": "wlts"
      }
    ]
}
```

LCCS

LCCS-WS (Land Cover Classification System Web Service) is a web service that uses the data model provided by the LCCS-DB package. The service aims to facilitate access to and visualization of **classes** and their **symbols** in each classification system, as well as the **mappings** performed between the classes of the classification systems, in order to simplify the joint analysis of the data.



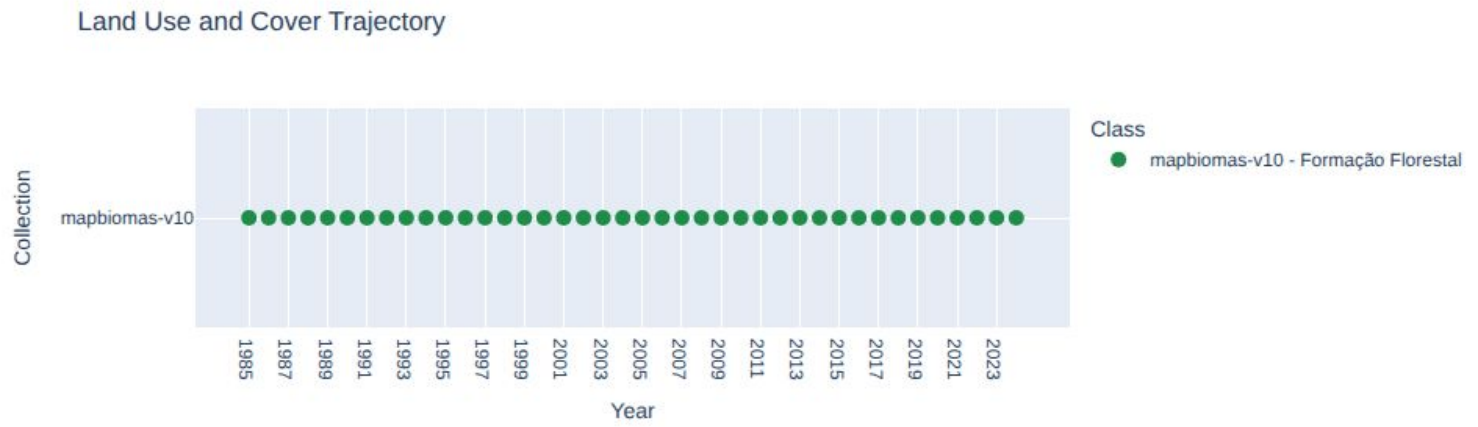
Clients in Python

It allows integration with the WLTS service and some packages such as pandas and geopandas.

It also offers support for creating applications and their use in interactive computing and through the command-line interface.

```
1  tj = service.tj(latitude=-12.0, longitude=-54.0, collections="mapbiomas-v10")
✓ 1.9s
```

```
1  service.plot(tj.df())
✓ 1.2s
```




```
lts.py) fabi@note-vaio:~/Documents/bdc/2026/wlts$ wlts-cli --url 'https://data.inpe.br/bdc/wlts/v1'
        trajectory \
        --latitude -12.0 \
        --longitude -54.0  \
        --start-date 2001 \
        --end-date 2012 \
        --collections mapbiomas-v10
Processing trajectory request ... ———————————————————————— 0:00:00
      Trajectory Results (Time: 0.86 seconds)
```

Class	Collection	Date	Point ID
Formação Florestal	mapbiomas-v10	2001	1
Formação Florestal	mapbiomas-v10	2002	1
Formação Florestal	mapbiomas-v10	2003	1
Formação Florestal	mapbiomas-v10	2004	1
Formação Florestal	mapbiomas-v10	2005	1
Formação Florestal	mapbiomas-v10	2006	1
Formação Florestal	mapbiomas-v10	2007	1
Formação Florestal	mapbiomas-v10	2008	1
Formação Florestal	mapbiomas-v10	2009	1
Formação Florestal	mapbiomas-v10	2010	1

```
pip install wlts.py
```

https://github.com/brazil-data-cube/wlts.py

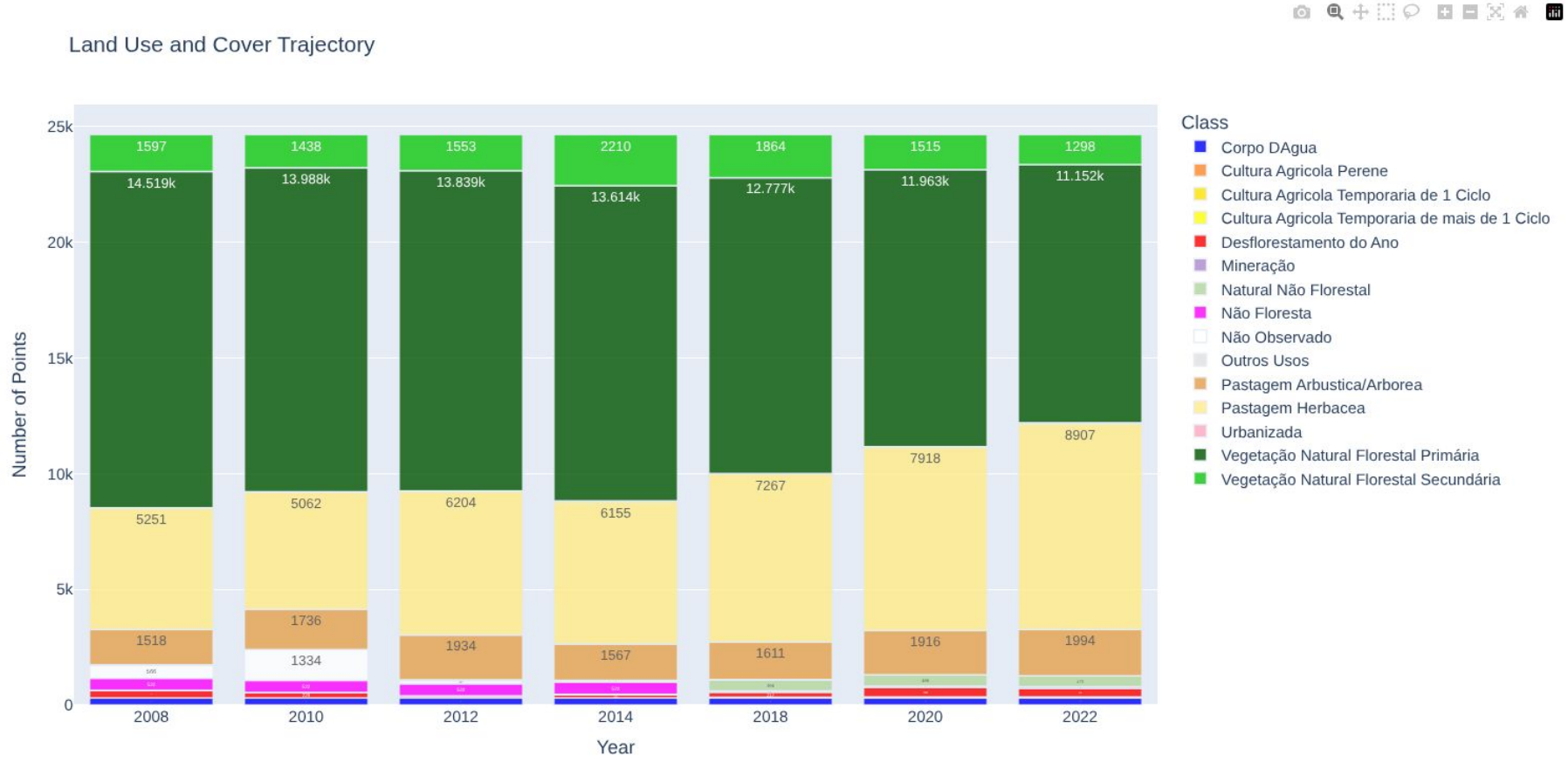
Integration: Terminal and Interactive Jupyter

```
1  service.plot(df, type="bar", marker_size=15, width=1500, height=750, font_size=15, textfont_size=13)
```

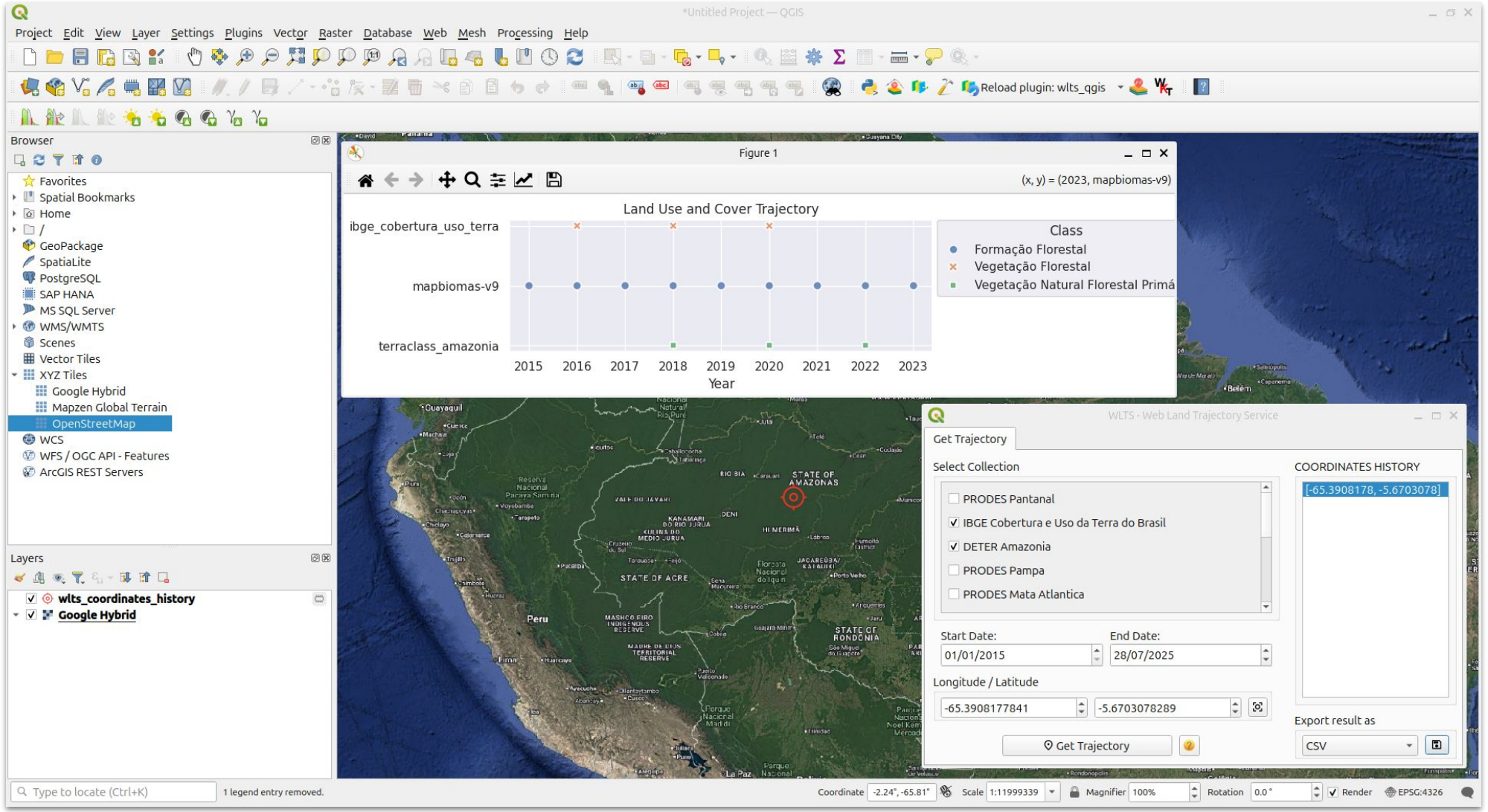
15] ✓ 0.8s

```
/home/fabi/.virtualenvs/wlts.py/lib/python3.11/site-packages/wlts/wlts.py:430: FutureWarning:

The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default an
```



Plugin QGIS

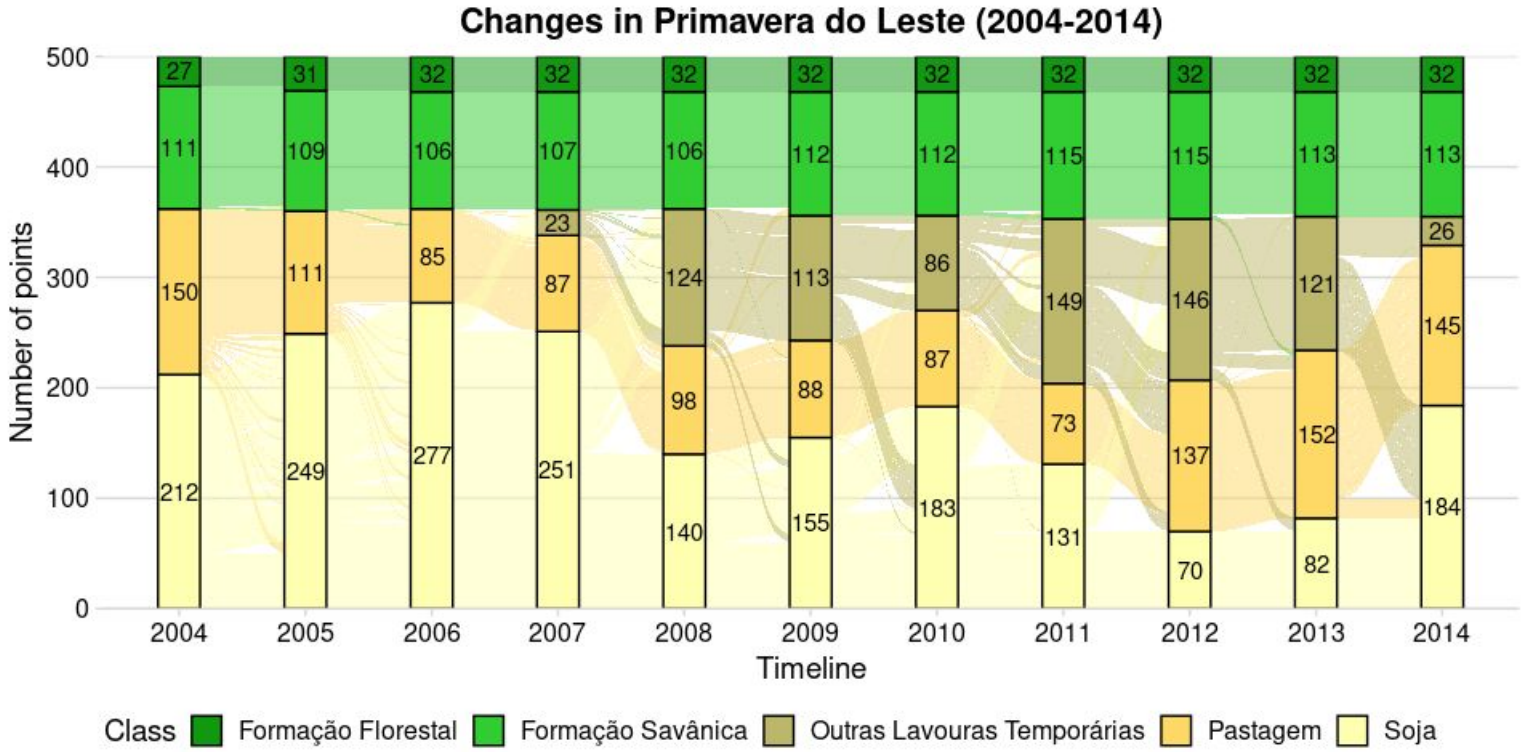


https://github.com/brazil-data-cube/wlts-qgis

Cliente em r: rwlts

RWLTS é um pacote R para interação com o WLTS.

A API do pacote é compatível com as funções e a nomenclatura fornecidas na biblioteca Python.




```
devtools::install_github("brazil-data-cube/rwlts"
                         )
```

https://github.com/brazil-data-cube/rwlts

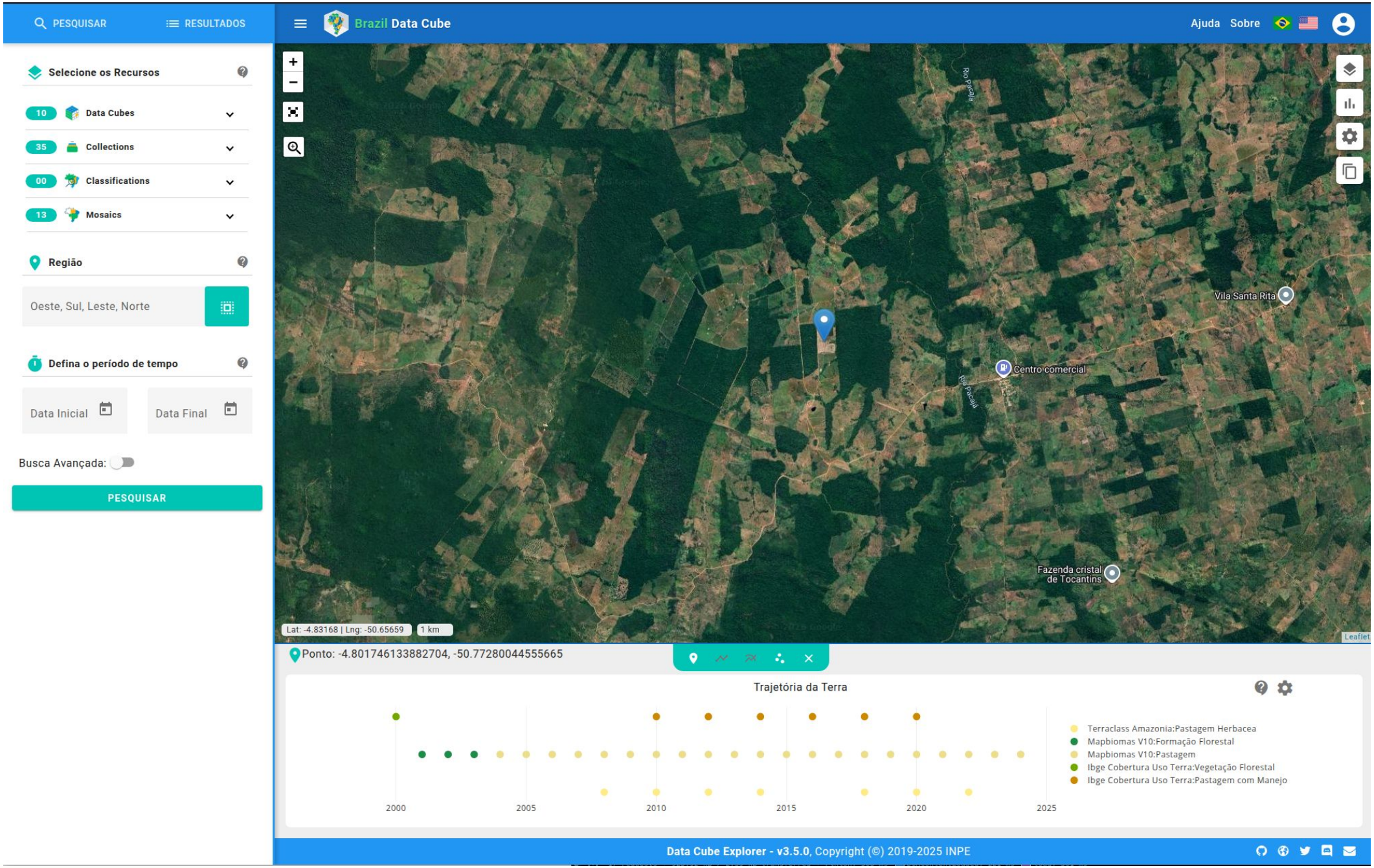
```
get_trajectory(wlts_bdc,
               latitude = -12,
               longitude = -54,
               collections = "mapbiomas_amazonia-v5",
               query_info = TRUE,
               config = httr::add_headers("x-api-key" = "change-me"))
```

BDC-Explorer

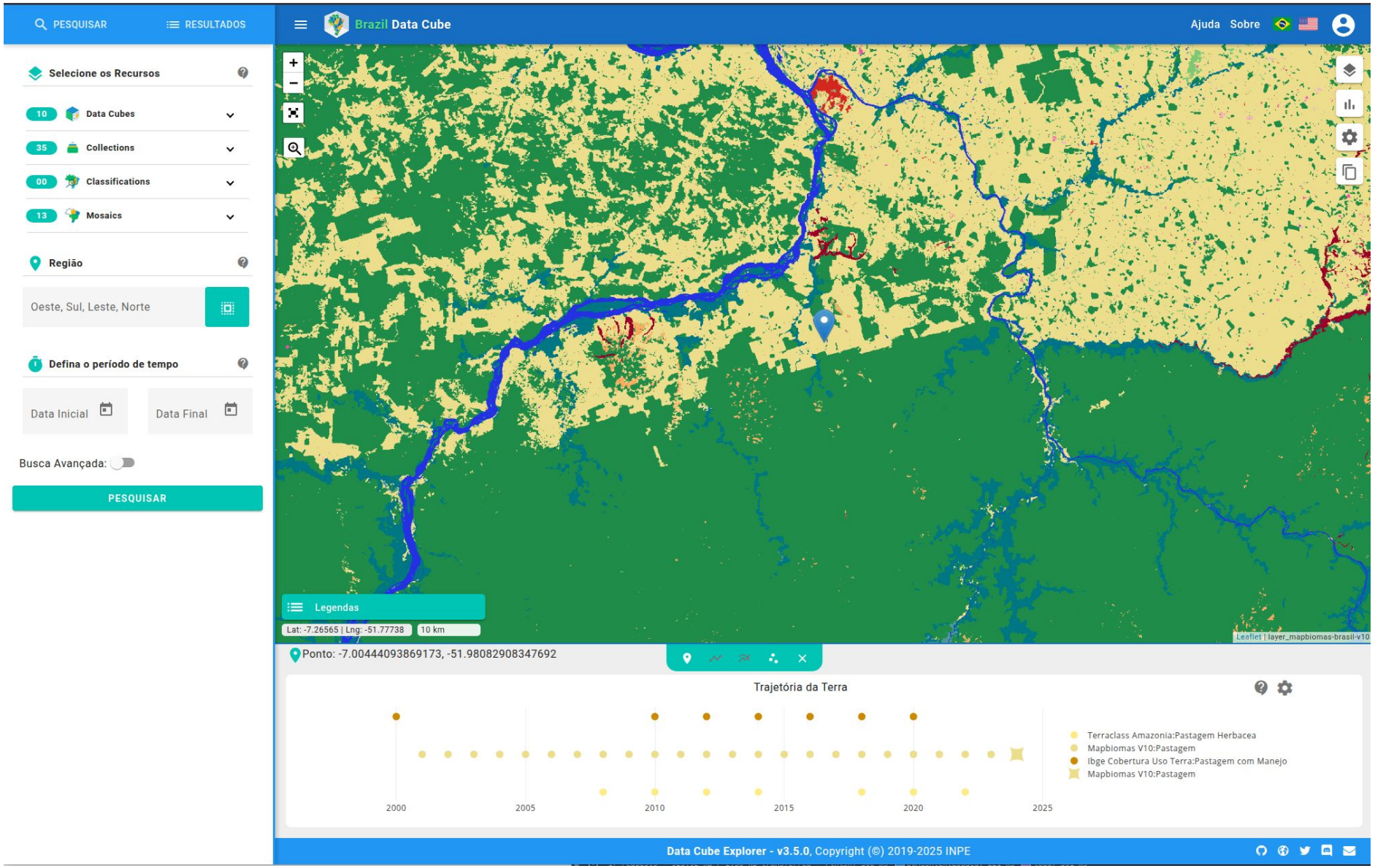
Web interface for WLTS

https://data.inpe.br/bdc/explorer

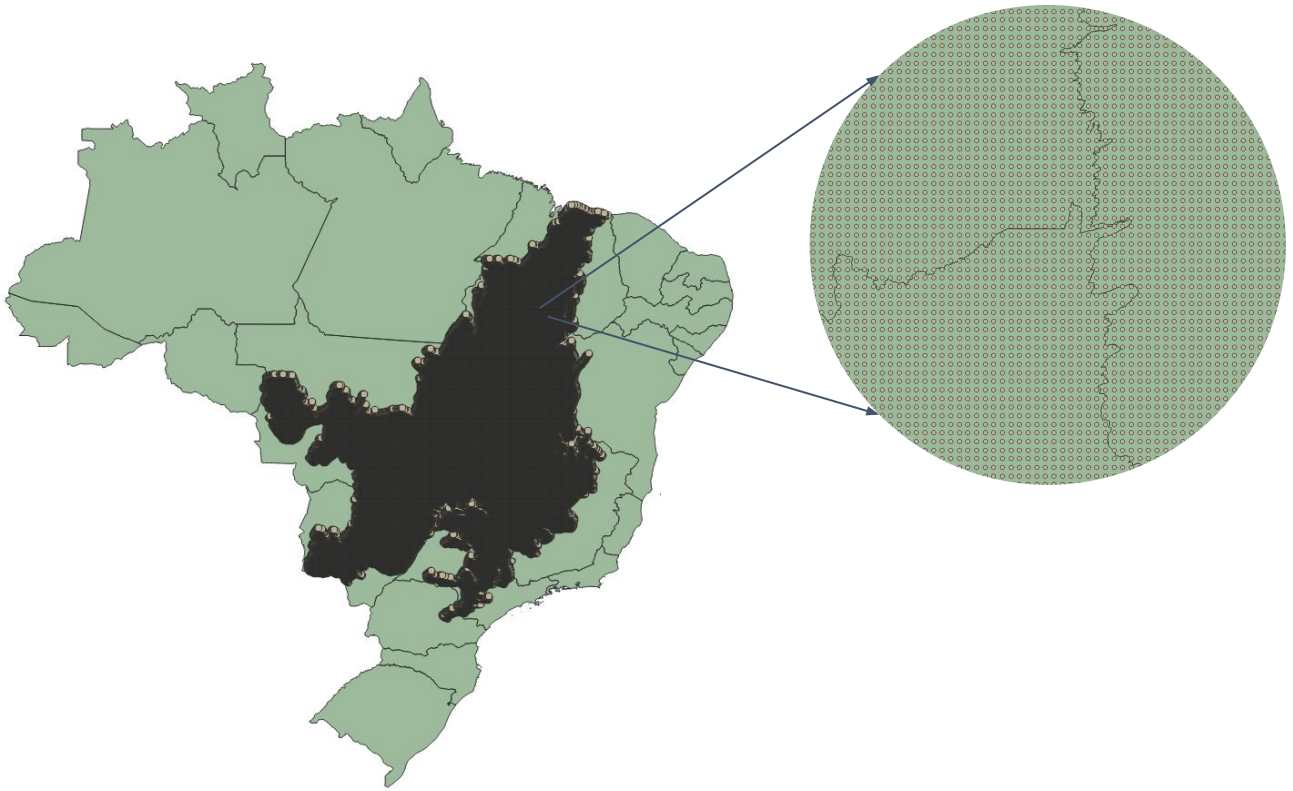



PESQUISAR
RESULTADOS
Brazil Data Cube
Ajuda
Sobre
Selecione os Recursos
10 Data Cubes
35 Collections
00 Classifications
13 Mosaics
Região
Oeste, Sul, Leste, Norte
Defina o período de tempo
Data Inicial
Data Final
Busca Avançada:
PESQUISAR
Vila Santa Rita
Centro comercial
Fazenda cristal de Tocantins
Lat: -4.83168 | Lng: -50.65659
1 km
Leaflet
Ponto: -4.801746133882704, -50.77280044555665
Trajetória da Terra
2000
2005
2010
2015
2020
2025
Terraclass Amazonia:Pastagem Herbacea
Mapbiomas V10:Formação Florestal
Mapbiomas V10:Pastagem
Ibge Cobertura Uso Terra:Vegetação Florestal
Ibge Cobertura Uso Terra:Pastagem com Manejo
Data Cube Explorer - v3.5.0, Copyright (©) 2019-2025 INPE

Aplicações Web: BDC-Explorer



Examples of Applications



Comparison between Collections

Class	Collection	Date
Vegetação Campestre Formação Savânica	ibge_cobertura_uso_terra mapbiomas-v10	2018 2018

Class	Collection	Date
Área Agrícola Cana	ibge_cobertura_uso_terra mapbiomas-v10	2018 2018

Extraction of training samples based on concordances between different LULC datasets (Lapig Pastagem, IBGE, Mapbiomas): 5 × 5 km grid of the Cerrado biome. With 85,026 samples.

Exemplos Aplicações

From the 85,026 samples, those in which there was no disagreement between the labels assigned by the three sources were selected. The resulting set had 48,850 points, from which the time series were extracted using the Landsat-8 data cube.

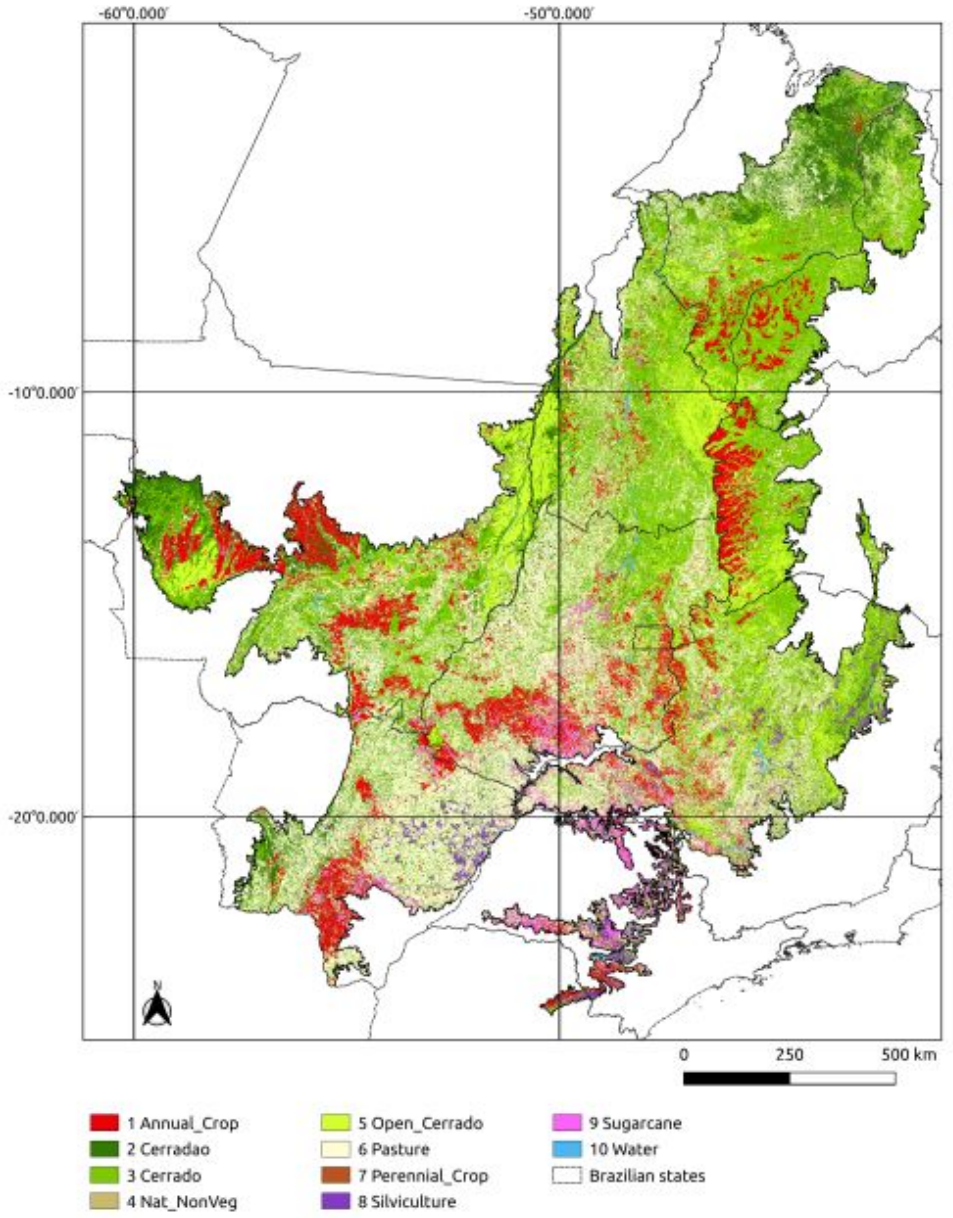
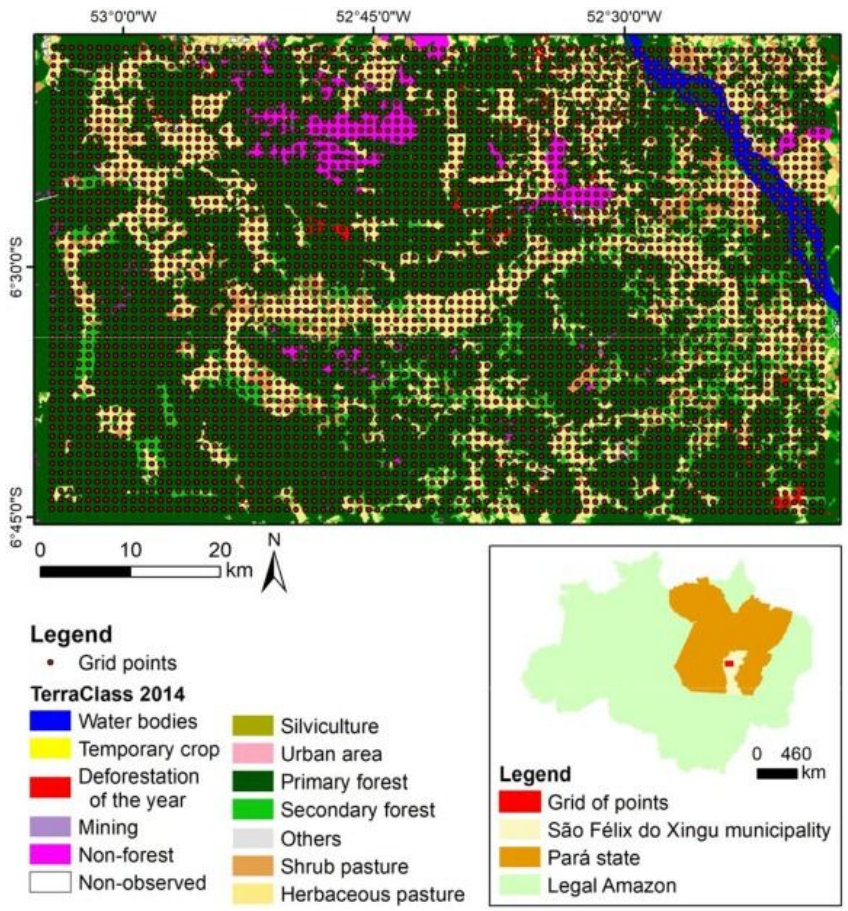
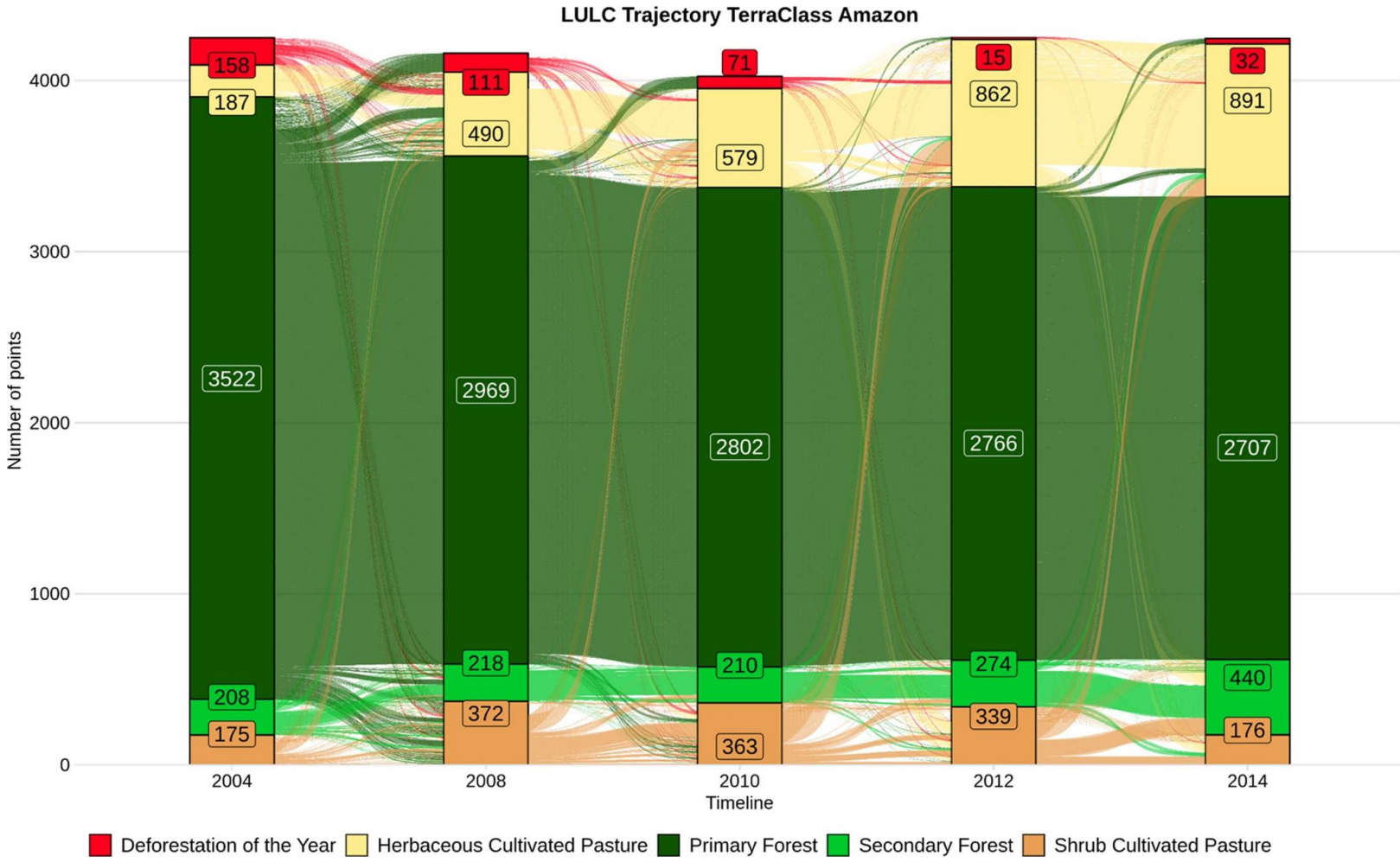


Figure 7. Cerrado land use and land cover map for 2018 (source: authors).

Fonte: Simoes et al. (2021)

WLTS: Case Study





WLTS: Case Study

Based on LULC trajectories, we calculated the agreement between **TerraClass** and **Mapbiomas**. The table presents the agreement analysis, considering the simplified legend. The overall agreement was approximately 85%.

Table 5

Agreement analysis between TerraClass (TC) and MapBiomas for 2014, considering 4472 points in the study area shown in Fig. 1. Overall agreement is 85.89% and the classes abbreviations are: Non observed (NO), Deforestation of the year (DY), Urban area (UA) and Non forest natural formation (NFNF)

		MapBiomas								
		Forest	Pasture	NO	Water	DY	Mining	Others	UA	NFNF
TC	Forest	2806	330	0	6	–	0	0	0	4
	Pasture	91	975	0	1	–	0	0	0	1
	NO	106	15	0	0	–	0	0	0	27
	Water	5	0	0	60	–	0	0	0	0
	DY	14	18	0	0	–	0	0	0	0
	Mining	0	3	0	1	–	0	0	0	0
	Others	2	5	0	1	–	0	0	0	0
	UA	0	1	0	0	–	0	0	0	0
	NFNF	0	0	0	0	–	0	0	0	0

References

Zioti, F.; Ferreira, K. R.; Queiroz, G. R.; Neves, A. K.; Carlos, F. M.; Souza, F. C.; Santos, L. A.; Simoes, R. E. O. A platform for land use and land cover data integration and trajectory analysis. International Journal of Applied Earth Observation and Geoinformation. V 106, P 102655, Feb 2022.

Fabiana Zioti et al. Um ambiente para acesso e análise de trajetórias de uso e cobertura da Terra. In: ANAIS DO XIX SIMPóSIO BRASILEIRO DE SENSORIAMENTO REMOTO, 2019, Santos. Anais eletrônicos..., INPE, 2019. Disponível em: <https://proceedings.science/sbsr/sbsr-2019/trabalhos/um-ambiente-para-acesso-e-analise-de-trajetorias-de-uso-e-cobertura-da-terra?lang=pt-br>.

Anjos, A. E. dos ., Vieira, F. Z., Ferreira, K. R., & Queiroz, G. R. de . (2025, September 23). WLTS-QGIS: uma extensão para recuperação de trajetórias de uso e cobertura da terra no ambiente QGIS. XXV Workshop de Computação Aplicada, INPE. https://doi.org/10.5281/zenodo.17187264

Ferreira, K.R.; Queiroz, G.R.; Vinhas, L.; Marujo, R.F.B.; Simoes, R.E.O.; Picoli, M.C.A.; Camara, G.; Cartaxo, R.; Gomes, V.C.F.; Santos, L.A.; Sanchez, A.H.; Arcanjo, J.S.; Fronza, J.G.; Noronha, C.A.; Costa, R.W.; Zaglia, M.C.; Zioti, F.; Korting, T.S.; Soares, A.R.; Chaves, M.E.D.; Fonseca, L.M.G. Earth Observation Data Cubes for Brazil: Requirements, Methodology and Products. Remote Sens. 2020, 12, 4033. https://doi.org/10.3390/rs12244033

Simoes, R., Camara, G., Queiroz, G., Souza, F., Andrade, P. R., Santos, L., Carvalho, A., & Ferreira, K. (2021). Satellite Image Time Series Analysis for Big Earth Observation Data. Remote Sensing, 13(13), 2428. https://doi.org/10.3390/rs13132428

Thank you!

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